

Self-Evaluation

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Major accomplishments

- ✦ After almost 10 years of teaching Chemical Engineering Thermodynamics and writing hundreds of pages of notes as well as assignments for my students, I completed a textbook which will be issued to the incoming class (free of charge) in the Fall of 2025. This textbook is the culmination of a ten-year journey—one shaped not only by countless hours of reflection and writing, but also by the insights, questions, and enthusiasm of the many students I have had the privilege to teach. I am deeply grateful to all the students who contributed along the way through thoughtful discussions, constructive feedback, and a shared commitment to understanding chemical engineering thermodynamics. Their curiosity and persistence inspired much of the content and clarity found in the textbook. I plan to dedicate this textbook to them—with sincere thanks and admiration—for making teaching not just a profession, but a calling filled with purpose and joy.
- ✦ A new version 12.0 of the **CHE Calculator**, the state-of-the-art computational tool used for completing assignments in all the courses I teach at UIC—Thermodynamics, Process Simulation, and Process Control—will also be issued in the Fall of 2025 to incoming classes of juniors and seniors. The CHE Calculator contains **six computational modules** continuously being enhanced to aid the students in learning the fundamentals of chemical engineering and applying them to real-world problems. The **CHE Calculator** has been and continues to be in continuous development for the last 10 years, with numerous contributions from my students, and covers all major areas of chemical engineering, with special emphasis on thermodynamics and process control.
- ✦ Co-authored a paper with Betül Bilgin that was presented at the ASEE conference (June 2025). The title of the paper was: ***CHE CALCULATOR: Empowering Students with Computational Tools for Real-World Engineering Challenges.***

Objectives for the upcoming academic year

- ✦ My primary objective for the upcoming academic year is to complete a new textbook on Chemical Process Control, which I have also been working on for the past ten years. Approximately five chapters of the textbook have been written, and three remain to be completed. The first edition of the textbook is scheduled to be released in the Spring of 2026.
- ✦ At the same time, I plan to begin compiling the hundreds of pages of notes I have written on the subject of Process Simulation with ASPEN Plus, in order to produce a textbook that future students can use both in my course and in their professional careers. The first edition of this textbook is scheduled to be released in the Fall of 2026.

Advancing the learning and career prospects of our students

- ✦ In my teaching, I strive to balance fundamental theory with real-world applications to enhance students' learning and increase their comfort level in solving real-world problems. Toward that goal, this year I organized a field trip to the Dresden nuclear facility in Morris, Illinois, where students experienced firsthand the reality of using nuclear fission to produce electric power. They had a chance to see the enhanced Rankine cycle for power production. Prior to the visit, they had the opportunity to use the **CHE Calculator** to simulate the Rankine cycle and see how thermodynamics can be used for the good of the people. In the previous year I organized a field trip to Novaspect, an Impact Partner of Emerson Automaton Solutions servicing industrial process controls. The students heard from senior executives and process control chemical engineers about the company philosophy, its services in process automation, digitalization, digital twins and cyber security.
- ✦ Additionally, I always try to bring engineers from the local chemical industry to give presentations on a variety of subjects including safety, solids handling, simulation, and industrial automation. These lectures reinforce students' learning and expand their horizons by allowing them to meet people from industry. I will continue doing that, as I pledge to work tirelessly to enhance the learning of my students and contribute to their ambition to become good chemical engineers who work for the betterment of the people.

Summary of self-evaluation

As a professor, but mostly as a human being, I try to demonstrate an exceptional dedication to both teaching and scholarship in chemical engineering, particularly in the field of thermodynamics, process simulation and control. Over the course of a decade I have meticulously developed comprehensive course materials, including multiple textbooks and computational tools, to enhance student learning and engagement.

My teaching philosophy strikes a thoughtful balance between rigorous theoretical foundations and real-world applications, enabling students to not only understand complex concepts but also to appreciate their practical relevance. This approach is evident in the integration of field trips, guest lectures from industry professionals, and computational tools like the **CHE Calculator**, all designed to broaden student horizons and prepare them for professional challenges.

I effortlessly strive for a consistent commitment to continuous improvement and dedication to mentoring future engineers who will contribute meaningfully to society. In summary, I try to embody the qualities of an educator and scholar, by combining intellectual rigor, pedagogical innovation, and genuine care for my students' success and professional growth.